

Steganography

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 Writer	© Samin Ahmed Chowdhury
 Vedio1	Infosec BD https://youtube.com/playlist?list=PL60oESahDerqJ4QOw3iPy9SmUEwtOxd2V&si=cKhfJys_7tPjpFjk
 Vedio2	https://youtu.be/7ln6yuM-OWw?si=i2LLNveZ_3AFZjpS

Topics :

What is Steganography

Details :

Steganography :

Steganography

- **Steganography:**

Steganography is process that is use for hiding information inside of **Image, audio , video or any digital object.**

Example:

My Bank Account number is :1234324

CMD :

`steghide`

To embed emb.txt in cvr.jpg :

```
steghide embed -cf cvr.jpg -ef emb.txt
```

To extract embedded data from stg.jpg:

```
steghide extract -sf stg.jpg
```

Tools :

- a.Xiao
- b.cryture
- c.steghide
- d.zsteg
- e.Openstego
- f.Foremost

<https://www.kali.org/tools/gobuster/>

My Favourite Tools or Command:

- a. strings ----- default command On Linux OS
- b. Stegsolve (java file)
- c. exiftool (Default On Linux OS)
- d. Binwalk (Default on Kali Linux OS)
- e. Sonic visualizer

resource : [0xrick Stego](#)

1. Use of Steghide tools
2. Use of zsteg tools
3. Use of exiftool tools
4. Use of stegoveritas
5. Use of Sonic Visualizer tools
6. How to solve cc: steganography room of Try Hack Me

```
sudo apt install steghide
```

all details about steghide 📌

```
steghide -help
```

```
unzip x.zip
```

 unzip ke zip kora

```
(foysal@kali)-[~/Downloads]
$ ls
Hacker404.ovpn  shadowCTF  spect  spect.zip  stegsolve.jar

(foysal@kali)-[~/Downloads]
$ cd spect
```

```
(foysal@kali)-[~/Downloads/spect]
$ ls
jpeg1.jpeg  jpeg2.jpeg  jpeg3.jpeg  png1.png  wav1.wav  wav2.wav

(foysal@kali)-[~/Downloads/spect]
$
```

PNG file we use zsteg :

```
gem install zsteg
```

```
zsteg -h
```

 details about zsteg

```
zsteg
```

 same

```
(samin@kali:~)$ zsteg
Usage: zsteg [options] filename.png [param_string]

-a, --all          try all known methods
-E, --extract NAME  extract specified payload, NAME is like '1b,rgb,lsb'

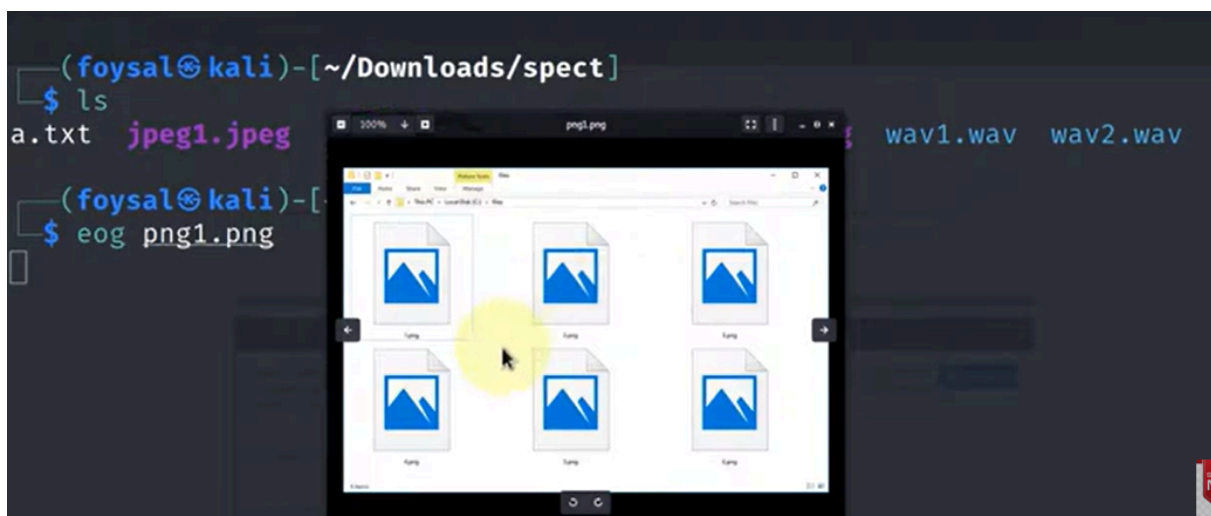
Iteration/extraction params:
-o, --order X      pixel iteration order (default: 'auto')
                   valid values: ALL,xy,yx,XY,YX,xY,xy,bY,...
-c, --channels X    channels (R/G/B/A) or any combination, comma separated
                   valid values: r,g,b,a,rg,bgr,rgba,r3g2b3,...
-b, --bits N        number of bits, single int value or '1,3,5' or range '1-8'
                   advanced: specify individual bits like '00001110' or '0x88'
--lsb              least significant bit comes first
--msb              most significant bit comes first
-P, --prime         analyze/extract only prime bytes/pixels
--shift N          prepend N zero bits
--invert           invert bits (XOR 0xff)
--pixel-align      pixel-align hidden data

Analysis params:
-l, --limit N       limit bytes checked, 0 = no limit (default: 256)
--[no-]file        use 'file' command to detect data type (default: YES)
--no-strings       disable ASCII strings finding (default: enabled)
-s, --strings X     ASCII strings find mode: first, all, longest, none
                   (default: first)
-n, --min-str-len X minimum string length (default: 8)

-v, --verbose       Run verbosely (can be used multiple times)
-q, --quiet         Silent any warnings (can be used multiple times)
-C, --[no-]color    Force (or disable) color output (default: auto)

PARAMS SHORTCUT
zsteg fname.png 2b,b,lsb,xy ==> --bits 2 --channel b --lsb --order xy
```

solving :



working cmd :

```
zsteg x.png
```

output "nootnoot" ar green mark kora part ta "Payload"

```
sudo apt install exiftool
```

answer :

****eda aktu set up korte hobe error issue when run**

```
stegoveritas -help
```

```

(foysal@kali)~/Downloads/spect]
$ stegoveritas -steghide jpeg2.jpeg
Running Module: SVImage
+-----+
| Image Format | Mode |
+-----+
| JPEG (ISO 10918) | RGB |
+-----+
Found something with StegHide: /home/foysal/Downloads/spect/results/steghide_c3d5df5a96b02d3f339a8b81891aeb4b.bin
Running Module: MultiHandler

```

```

(foysal@kali)~/Downloads/spect]
$ cd results

(foysal@kali)~/Downloads/spect/results]
$ ls
keepers  steghide_c3d5df5a96b02d3f339a8b81891aeb4b.bin

(foysal@kali)~/Downloads/spect/results]
$ cd results

```

```

(foysal@kali)~/Downloads/spect]
$ cd results

(foysal@kali)~/Downloads/spect/results]
$ ls
keepers  steghide_c3d5df5a96b02d3f339a8b81891aeb4b.bin

(foysal@kali)~/Downloads/spect/results]
$ cat steghide_c3d5df5a96b02d3f339a8b81891aeb4b.bin
kekekkek
(foysal@kali)~/Downloads/spect/results]
$

```

cat dey bair korte hoi

Spectrogram :

ae part ta dekhle hobe

mainly audio ke ney kaj kore

*using **sonic visualizer**

flag Solve :

```

(foysal@kali)-[~/Downloads]
$ ls
a.txt          exam2.wav      qrcode.png    shadowCTF
exam1.jpeg     Hacker404.ovpn results        stegsolve.jar

(foysal@kali)-[~/Downloads]
$ file exam1.jpeg
exam1.jpeg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1,
segment length 16, Exif Standard: [TIFF image data, big-endian, direntries
=5, name=password=admin, xresolution=90, yresolution=98, resolutionunit=1],
baseline, precision 8, 213x160, components 3

(foysal@kali)-[~/Downloads]
$

```

```

(foysal@kali)-[~/Downloads]
$ exiftool exam1.jpeg
ExifTool Version Number      : 12.14
File Name                    : exam1.jpeg
Directory                    : .
File Size                    : 8.6 KiB
File Modification Date/Time   : 2021:01:26 12:53:52-08:00
File Access Date/Time        : 2021:01:26 12:54:04-08:00
File Inode Change Date/Time   : 2021:01:26 12:53:52-08:00
File Permissions              : rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                  : 1.01
Exif Byte Order               : Big-endian (Motorola, MM)

```

```

File Size                    : 8.6 KiB
File Modification Date/Time   : 2021:01:26 12:53:52-08:00
File Access Date/Time        : 2021:01:26 12:54:04-08:00
File Inode Change Date/Time   : 2021:01:26 12:53:52-08:00
File Permissions              : rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                  : 1.01
Exif Byte Order               : Big-endian (Motorola, MM)
Document Name                 : password=admin
X Resolution                  : 1
Y Resolution                  : 1
Resolution Unit               : None
Y Cb Cr Positioning           : Centered
Image Width                   : 213

```

```

(foysal@kali)-[~/Downloads]
$ steghide extract -sf exam1.jpeg
Enter passphrase:
the file "a.txt" does already exist. overwrite ? (y/n) Y
steghide: did not write to file "a.txt".

(foysal@kali)-[~/Downloads]
$ clear

```

```

(foysal@kali)-[~/Downloads]
$ ls
a.txt          exam2.wav      qrcode.png    shadowCTF
exam1.jpeg     Hacker404.ovpn results        stegsolve.jar

(foysal@kali)-[~/Downloads]
$ cat a.txt
the key is: sUpErkeykey

(foysal@kali)-[~/Downloads]
$

```

Flag : Music Stego :

Sonic

```

(foysal@kali)-[~/Downloads]
$ ls
binaries.zip  Hacker404.ovpn  musicalstego.wav  shadowCTF  stegsolve.jar

(foysal@kali)-[~/Downloads]
$ steghide extract -sf musicalstego.wav

```

```

(foysal@kali)-[~/Downloads]
$ ls
binaries.zip  Hacker404.ovpn  musicalstego.wav  shadowCTF  stegsolve.jar

(foysal@kali)-[~/Downloads]
$ steghide extract -sf musicalstego.wav
Enter passphrase:
wrote extracted data to "secret".

(foysal@kali)-[~/Downloads]
$

```

```

(foysal@kali)-[~/Downloads]
$ ls
binaries.zip  Hacker404.ovpn  musicalstego.wav  shadowCTF  stegsolve.jar
secret

(foysal@kali)-[~/Downloads]
$ file secret
secret: ASCII text

(foysal@kali)-[~/Downloads]
$ cat secret
T!HM{f0und_m3!}

(foysal@kali)-[~/Downloads]
$

```

Binwalk, Stegsolve :

1. Binwalk:

```
binwalk -e target_file
```

|

2. stegsolver:

```
java -jar stegsolve.jar
```

```
(samin@kali)~$ binwalk
Binwalk v2.4.3
Original author: Craig Heffner, ReFirmLabs
https://github.com/OSPGBinwalk

Usage: binwalk [OPTIONS] [FILE1] [FILE2] [FILE3] ...

Disassembly Scan Options:
-Y, --disasm          Identify the CPU architecture of a file using the capstone disassembler
-T, --mins=<int>      Number of consecutive instructions to be considered valid (default: 500)
-k, --continue        Don't stop at the first match

Signature Scan Options:
-B, --signature       Scan target file(s) for common file signatures
-R, --raw=<str>       Scan target file(s) for the specified sequence of bytes
-A, --opcodes         Scan target file(s) for common executable opcode signatures
-m, --magic=<file>    Specify a custom magic file to use
-b, --dumb            Disable smart signature keywords
-l, --invalid         Show results marked as invalid
-x, --exclude=<str>   Exclude results that match <str>
-y, --include=<str>   Only show results that match <str>

Extraction Options:
-e, --extract         Automatically extract known file types
-D, --dd=<type[:cmd]> Extract <type> signatures (regular expression), give the files an extension of <ext>, and execute <cmd>
-M, --matryoshka      Recursively scan extracted files
-d, --depth=<int>     Limit matryoshka recursion depth (default: 8 levels deep)
-C, --directory=<str> Extract files/folders to a custom directory (default: current working directory)
-j, --size=<int>      Limit the size of each extracted file
-n, --count=<int>     Limit the number of extracted files
-O, --run-as=<str>    Execute external extraction utilities with the specified user's privileges
-l, --preserve-symlinks Do not sanitize extracted symlinks that point outside the extraction directory (dangerous)
```

```
$ binwalk -e hell.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, JFIF standard 1.02
30	0x1E	TIFF image data, big-endian, offset of first image directory: 8
265845	0x40E75	Zip archive data, at least v2.0 to extract, uncompressed size: 69, name: hello_there.txt
266099	0x40F73	End of Zip archive, footer length: 22

```
(foysal@kali)~$ ls
dark.png      hell.jpg      shadowCTF
Hacker404.ovpn _hell.jpg.extracted stegsolve.jar

(foysal@kali)~$ cd _hell.jpg.extracted

(foysal@kali)~/Downloads/_hell.jpg.extracted$ ls
40E75.zip  hello_there.txt

(foysal@kali)~/Downloads/_hell.jpg.extracted$ clear
```

```

(foysal@kali)-[~/Downloads/_hell.jpg.extracted]
$ ls
40E75.zip  hello_there.txt

(foysal@kali)-[~/Downloads/_hell.jpg.extracted]
$ cat hello_there.txt
Thank you for extracting me, you are the best!

THM{y0u_w4lk_m3_0u7}

(foysal@kali)-[~/Downloads/_hell.jpg.extracted]
$

```

----- Multi Level Task in Steganography -----

File Name : trollcat.PNG
Flag Format : trollcat
Source : trollcat CTF Contest
Hints: My social media can help to find password
Points: 500
Author :mr grep

Stegcracker : pass cracker

----- Stegcracker -----
| I

Useages : Use for cracking .jpg , .mp3 , mp4 format file

Assume that you have .jpg format password protect file and inside of image file has important information.

Now , you want to extract information from your target file , what can you do???

Well , we need password of target file for extracting information .For that , we will use stegcracker :

Install Command :

```
pip3 install stegcracker
```

Updating Command:

```
pip3 install stegcracker -U --force-reinstall
```

Useages : Use for cracking .jpg , .mp3 , mp4 format file

Assume that you have .jpg format password protect file and inside of image file has important information.

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Install Command :

```
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```

Updating Command:

```
pip3 install stegcracker -U --force-reinstall
```

Stegcraker Command:

```
stegcracker target file wordlist
```

Example Command:

```
stegcracker image.jpg /usr/share/wordlist/rockyou.txt
```